

02_METHOD

Method Justification Statement

1. Refined Research Question

Primary Research Question

To what extent does digital transformation influence the quality and efficiency of higher education at the Universidad Nacional Mayor de San Marcos, and through which organizational and technological mechanisms does this influence occur?

Supporting Research Questions

1. How do the dimensions of digital transformation influence educational quality at UNMSM?
2. How does digital transformation affect the efficiency of academic and administrative processes at UNMSM?
3. Which organizational, technological, and human factors facilitate or hinder digital transformation at UNMSM?
4. What strategic actions should be implemented to strengthen digital transformation and improve educational quality and institutional efficiency?

2. Candidate Research Methods

Method 1 – Quantitative Survey Research.

Method 2 – Qualitative Case Study.

Method 3 – Mixed-Methods Case Study.

3. Method-Fit Matrix

Criterion	Quant.	Qual.	Mixed	Score	Selected
Alignment with research question	4	4	5	5	✓
Ability to measure impact	5	2	5	5	✓
Contextual understanding	2	5	5	5	✓
Practical applicability	4	4	5	5	✓
Methodological robustness	4	4	5	5	✓
Total	19	19	25	25	Mixed

4. Methodological Foundation

Creswell (2023): According to Creswell, mixed methods research is appropriate when complex problems require both quantitative measurement and qualitative explanation. Digital transformation affects technological, organizational and educational dimensions simultaneously; therefore, integrating numerical evidence with contextual interpretation provides a more complete understanding.

Yin (2024): Yin argues that case study research is the preferred strategy for investigating contemporary phenomena within their real-world context. Since digital transformation is embedded in the institutional environment of UNMSM, the case study design allows an in-depth analysis of interactions among governance, technology, culture, and educational processes.

Saunders, Lewis & Thornhill (2023): The Research Onion framework proposes that methodological decisions should derive from the research objectives. The present study combines a pragmatic philosophy, a case study strategy, and mixed methods to obtain rigorous and decision-oriented evidence.

Morgan (2024): Morgan identifies pragmatism as the philosophical foundation most compatible with mixed methods because it prioritizes the research question and the practical usefulness of knowledge. This perspective supports integrating multiple forms of evidence to solve institutional problems.

Tashakkori & Teddlie: These authors argue that triangulation and integration of quantitative and qualitative evidence increase credibility, validity, and explanatory power. Their framework supports combining surveys, interviews, institutional documents, and performance indicators in a coherent analytical design.

5. Justification of the Selected Method

The Mixed-Methods Case Study provides the strongest alignment with the research objectives because it measures the impact of digital transformation while explaining the organizational mechanisms that generate changes in educational quality and institutional efficiency. This design ensures methodological rigor, contextual depth, and practical relevance for evidence-based decision-making at UNMSM.

6. Expected Methodological Contribution

The proposed methodology contributes a replicable framework for evaluating digital transformation in public universities. It integrates statistical evidence, organizational analysis, and stakeholder perspectives to generate recommendations that can strengthen educational quality and institutional efficiency in Peru and similar emerging contexts.